Data 698 – Research Methods and Study Designs

Case Study Design: Autism Tweet Sentiment Analysis

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1. Abstract

With the help of sentiment analysis, this case study aims to help the autism community organize their autism awareness events efficiently. Using Twitter and past autism awareness event data, sentiment analysis will be done to determine the impact of each event on people's sentiment toward autism. The sentiment will be measured from tweets related to autism around the time of each event. The resulting outcomes will be a presentation to the autism community and a research paper. The presentation will communicate the results to the community and help them determine what type of events best achieve their goals. The information from the research will help these organizations better utilize their resources and achieve their goal more efficiently. The research paper, along with the code repository, will help further expand the research in the future. This research will be done with a team of two data scientists, two data labelers, a graphic designer, and a manager to organize everyone. The team will use resources such as cloud computing and multiple software to complete the research.

2. Introduction

This case study aims to determine what event and promotion strategies best help the autism community promote and better bring awareness to autism. The study will examine organized autism awareness events and determine which increased or decreased positive sentiment toward autism. Twitter is an excellent data source for analyzing public sentiment on diverse topics [25]. In the case of autism, tweets with autism-related words will be used to extract data on how particular autism awareness events affect people's sentiments. Analyzing the sentiment of autism tweets before and after autism awareness events will help determine how the different events affect people's sentiment toward autism. This analysis will look at the sentiment on a national scale. The focus would be on more noteworthy events where many resources would be utilized. Resources could be saved if any events with insignificant public sentiment changes are found. These resources could be public donations or government grants, which would be best not wasted. It would be ideal to utilize those resources to their best potential. Groups such as the Autism Society could benefit from adequately using such resources [3]. It would be a waste of resources to keep hosting a large event without regarding how significant their effects are. The optimized use of resources to make the public aware of autism would also help them attain resources.

This research will require using historical tweets and autism awareness event data. Using tweet data from the Twitter application programming interface (API) [25], the case study will look at tweets before and after the organized autism awareness event. The study would also require data on when autism awareness events occurred in the past. The websites of different organizations which contain information about past events would be used to gather information [14]. This case study will be exploratory as it would only determine which events in the past have led to significant positive sentiment toward autism. It would require further in-depth research to determine what factors about the events are affecting the sentiment. The significance referred to in the analysis would be determined relative to all the different events in the past.

The product of this case study will be a presentation and a research paper. The resulting analysis of how well the events in the past helped bring positive awareness to autism will help determine what awareness events the organizations should plan. The information from the analysis would benefit all the organizations working toward bringing positive awareness of autism, such as the Autism Society [3]. A presentation will be created for the organizations to understand the analysis and better plan their future events. In addition, the presentation should hopefully result in better use of donations and other financial resources provided to these organizations. The study will also help make these organizations more efficient in spreading factual information about autism to counteract the vast amount of online misinformation. The research paper would communicate how the case study was done and allow others to expand on it. This analysis could be expanded to places outside the United States to help third-world countries. There is still an issue where this study will not help the marginalized minority on Twitter, as the tweets will not best represent them. Overall, this analysis would help determine which events help bring positive awareness of autism to most of the public and allow for further research.

3. Data and Data Analysis

This case study utilizes two different datasets to conduct exploratory analysis. The first dataset is the historical Twitter data from the Twitter API. The second dataset is data about past autism awareness events.

3.1 Datasets

Both datasets have various sources with varying structures and methods of retrieval. Therefore, the process of preparing the data for analysis will be different for each dataset.

3.1.1 Autism Tweet Data

Retrieving the Twitter data will be straightforward because it is structured and organized by the Twitter API [25]. However, the process of accessing this interface is complicated. Standard access is limited in functionality but is available to the public [1]. However, the user must have academic research access to access the historical tweets beyond the past week [1]. The process of obtaining this access requires the user to fit specific criteria, and an application about the use case of the interface will need to be submitted. The case study can only continue once the request is approved. The user gains access to about ten million monthly tweets with academic research access.

The Tweepy library [24] for Python will be used to retrieve the tweets. The library requires authentication information such as the bearer token provided by the interface. Once the access to the interface is set up, the user can use the Tweepy library to hit specific endpoints with a query statement. For this case study, the query string would request tweets containing autism-related words with a language filter to restrict the tweets to only English. The list of autism-related words will be sourced from a glossary of autism terminology [8]. This list includes words such as autism, autistic disorder, Asperger's syndrome, and much more. The final list of terms used will be included in the code repository. Using the correct query string and hitting the endpoint for the historical tweet search [7] will result in millions of tweets being transferred back to the user. Next, the data stream is paginated and will require iterating through the pages to get all the tweets. The interface provides a variety of fields, but the case study only requires the created time, tweet string, and language field. This functionality allows for dimensionality reduction. Specifying which fields are desired would result in only the desired data being returned. Language is the only metadata that will be used. Locally caching the fetched data during the pagination process will help prevent memory overflow. One caching method is storing the data directly to a file as the data is being fetched. This method will allow the user to use the cached data rather than hitting the endpoint each time. The main concern with storing the data in a file is ensuring the data is formatted correctly and no information is lost due to encoding issues. The data will be stored as comma-separated values with three columns for each of the different fields.

The clean-up process can begin now that the data is cached and accessible locally. The data and different metadata associated with the tweets were stored and structured for easy retrieval. However, these tweets will require further clean-up to be used during the analysis process. First, the data will be read into a Dask data frame [19] to start the clean-up process. Next, the tweets will be processed using the Natural Language Toolkit (NLTK) [11] to clean up strings using regular expression and Dask data frame mapping to filter out unnecessary words and characters. The resulting strings will be tweets without hashtags, Twitter handles, and other symbols or emojis that will not be useful to the sentiment reading library. Most of the clean-up process helps remove ethical and privacy concerns as data points like hashtags and handles allow direct connection back to conversation or users. Therefore, eliminating those data points helps reduce ethical and privacy concerns. In future research, with the consent of users, hashtags and handles can be used to determine how a specific person or group of people changed with different autism events. Additionally, in the future, emojis could be translated into text using an emoji parsing library and used in the analysis. However, it will be removed now as TextBlob does not support them well. The overall clean-up process will result in tweets ready to be further preprocessed before the sentiment analysis.

3.1.2 Past Autism Awareness Event Data

Unlike the tweets, the autism awareness event data are unstructured and will require web scraping to pull together a dataset. Web scraping [28] is the process of extracting data from a website by reading the HTML content of the website. The most comprehensive data source for past autism awareness events is the Interagency Autism Coordinating Committee website [14]. Therefore, this website will be scraped to create the dataset. This source provides lists grouped by year for different events in the past. The HTML of the page can be downloaded using a library such as Requests [18]. The HTML contains data on each event with information on the name, hosting organization, and date. Some of the recent events also contain the times of the event along with the date. This granular information could be used further in the analysis process. Therefore, the time values will be saved as well.

Once the dataset is extracted from the HTML, it will require labeling to determine the event type. The labeling process will begin with the data scientist creating a straightforward labeling method that uses a dictionary where labels are associated with a list of different words with weights attached to each word in the list. The label list includes conferences, summits, meetings, training, and webinar. The data scientist creates the dictionary and attaches the weights to the words. This dictionary is stored in a file allowing for repeatability. In addition, the reasoning for the weights first discussed and assigned to each word is stored, ensuring consistency and repeatability.

Given the dictionary, the function iterates through all the events and labels events based on the following method. For each event name, words from the name are matched with words associated with each label, and the weights are summed up for each word found. The sums will vary per label as the weight of a word will be different for each label. If the word from the event name is missing in the list, then the weight is zero. For each word found, the weight values are summed up per label, and the label with the highest value is assigned to the event. This method can be adjusted iteratively to improve the accuracy. The iterative process for improving the accuracy starts with the data scientists creating a list of 50 to 100 random event names and labeling them manually with the reasoning stored. Then, adjusting the weights for the words in the dictionary until the accuracy reaches around 80%; trying to make it 100% accurate will only lead to overfitting. Additionally, accurately labeling 80% of the dataset will reduce the workload for the labelers and the data scientists when fixing inconsistencies. In the future, the weight adjusting part can also be automated using the term frequency-inverse document frequency [22] with the labeled dataset. The labelers will be used to fix any remaining inconsistencies. After the automated process, labelers will mend the automatically chosen labels and provide a reason for mending the label. Both labelers will go through all the events, and their reasoning will allow for repeatability and consistency in future research. The reasoning and the new label will be stored as separate columns per labeler. The data scientists will resolve conflicts between the labeler's decisions by discussing the conflicts with each other and storing the new label and the reasoning used to resolve the conflict. Conflicts include situations where one labeler mends while another does not and situations where the labelers do not agree. This method increases the complexity of the process, but it should help eliminate the consistency and repeatability concerns. The final labels consist of the ones confirmed by the data scientists while resolving the conflict, the unamended labels, and the agreed and mended labels by the labelers. Once labeled, further data clean-up will be required to prepare the dataset for analysis. This data set will be stored as comma-separated values as well.

The clean-up process for this dataset will be more cumbersome than the tweets. The different values in the dataset can be passed through a clean-up process to remove unwanted characters, but not much else can be done. Also, due to the nature of the date format and some missing time, the date/time clean-up process will have to accommodate all the different formats on the website. Once the clean-up is complete, the dataset should be ready for analysis.

3.2 Data Analysis

Sentiment analysis will be the primary analysis done on the datasets. The sentiment analysis process is simplified using the TextBlob library [21]. However, before analyzing the sentiment of the tweets, the data will need to be further preprocessed.

3.2.1 Preprocessing

Due to the extensive clean-up process, this preprocessing step is simplified. The TextBlob library contains many useful functions that can help improve the sentiment analysis results. These features include changing word inflection, lemmatization, and spelling correction. To avoid altering the content of the tweets drastically, the only preprocessing will be tokenization and spelling correction. The results from the preprocessing will be stored as a TextBlob object. This process will be immediately before each tweet's sentiment is analyzed. Therefore, it will be done during the analysis process, and the results will be used immediately to analyze sentiment. Both datasets will be kept separate until after the sentiment analysis. Then, they will be joined when analyzing the potential effect of the events on the tweets.

3.2.2 Sentiment Analysis

As stated earlier, the sentiment analysis will be done using the TextBlob library [21]. The library utilizes lexicon-based or rule-based sentiment analysis. Rules-based sentiment analysis assigns a weight to all the words and determines the polarity using those weights. Given the tweets, the TextBlob library provides back value for polarity and subjectivity. The polarity values are numeric and range from negative one to positive one. The polarity determines if the tweet is positive (greater than zero), negative (less than zero), or neutral (zero). The subjectivity value ranges from zero to one; zero is objective while one is highly subjective. The subjectivity score determines whether the tweet is objective or subjective. With that, the polarity score is calculated for all tweets. The results are stored in a column in the table containing the tweets.

This project will ignore the subjectivity score as it does not add value to the analysis. The subjectivity score tries to quantify how much of the string is an opinion or a fact. The primary output of this study only focuses on whether an autism awareness event affects the sentiment of tweets containing autism-related words. These changes include changing from negative to positive or positive to negative or not changing. Therefore, determining if the strings are opinions or facts will not add value to the goal. Additionally, determining if a string is a fact or opinion with a Lexicon-based sentiment analysis will introduce further complexity and potentially more errors. This data point adds unnecessary complexity to the case study for the primary goal. The subjectivity score can be used later to build on this case study to determine if an event helps lead to a more significant number of tweets with factual information about autism. Further analysis with subjectivity will require a greater focus on validating and correcting the potential errors, similar to how it will be done with polarity.

3.2.3 Sentiment Validation

Before continuing with the analysis, the polarity scores must be validated. This process will require a similar system to how the second dataset was validated, involving labelers and data scientists. This process is slightly different as it will first use a separate library with lexicon-based sentiment analysis to confirm the results. The TextBlob library is robust, but it is known to have problems detecting negative sentiment. Therefore, the VADER [26] sentiment analysis tool in NLTK [11

] will be used to validate the initial polarity scores. The VADER tool is known to handle negative sentiments better. To validate the polarity scores, all the cleaned-up tweet strings will be processed through the VADER tool to extract the new polarity score. The output format of the new library is different from TextBlob as it presents four different values. These values include a "pos" score, a "neg" score, a "neu" score, and a compound score. Each of the three scores other than compound range from zero to one and respectively determine how confident the model predicts that the string is positive, negative, and neutral; zero is not confident, and one is very confident. The compound score is similar to the score from TextBlob. The threshold for positive is 0.05 and above. For negative, the threshold is -0.05 and below. Neutral falls between 0.05 and -0.05. Only the compound score will be used as that is the only score that can be compared with TextBlob. The compound scores will be placed in a separate column in the same data frame holding the TextBlob polarity scores and tweets. A script will be created which flags the rows with conflicting polarity scores. It will not check the exact numeric score but the categorical label that the score represents. For example, the rows will be flagged if one is positive while another is negative or neutral. This process should reduce the number of tweets that need to be manually validated from millions to tens of thousands or thousands.

Once the rows are highlighted, the data labelers and the data scientists will follow the same system used for the second dataset to resolve the conflicts. Each data labeler will look through the flagged rows and pick either the TextBlob or the VADER score depending on what they think better represent the sentiment and record their reasoning in a separate column. They will not be able to create a new score. If the data labeler's choices conflict, then the two data scientists will discuss and pick one of the options and again record their reasoning. With that, all the scores should be validated. The validated scores will be in a separate column and used for further analysis.

3.2.4 Analysis

Once the sentiment of the tweets is analyzed and validated, the events are then connected to the tweets using the date and time values. The polarity of the tweets the day before is compared to the polarity during the day and the day after. The exact time of the event will be used in the case of the events with the time data. In the case of events with time data, the polarity of the tweets 24 hours before the event is compared to the polarity 24 hours after the event's start time. The analysis time range is increased up to a week as well. If any events are close, the time frames are set up to prevent collisions. If two or more events occur during the same day, the events are grouped and analyzed as one rather than multiple events. The combination of events is due to the inability to discern which event caused the sentiment change.

Overall, the resulting dataset will contain rows for each event or combined events with columns for the average polarity 24 hours before the event, average polarity 24 hours after the event, the average polarity a week before the event, if applicable, the average polarity a week after the event, if applicable, and the difference between polarity before and after the different timeframe. The change in polarity of tweets over time will be graphed on a line graph. Polarity will be on the y-axis with the date of each tweet on the x-axis. The graph will be generated with all tweets and then aggregated at different time lengths, including intervals like a day, week, month, and year. The aggregated results would also include separate graphs with each volume, average change in sentiment, the median change in sentiment, and extrema for change in sentiment on the y axis. These graphs would show how the sentiments and volume of tweets changed over time and how the volume of autism-related tweets changed over time.

The most important descriptive statistics will be generated by aggregating the data by event type. Once aggregated by event type, faceted boxplots [6], which allow for comparison, will also be created, showing the overall change in polarity for each event type. Event types will be on the x-axis and the change in polarity on the y-axis. The boxplots will also show the median, maximum change, minimum change, and other box plot characteristics for each event type. These graphs will allow the community to see what event types are better than others. In the case of combined events, due to their temporal proximity, the statistics from the combined event will be used by each event type included in the combined event. For example, suppose a combined event consists of a meeting and a conference. In that case, the change in polarity associated with the combined events will be added to the conference event type and meeting event type statistics. Additionally, another graph will be created where the combined events are ignored, and only the uncombined events will be aggregated by event type. This graph will allow people to see how the combined events affect the data points. These graphs will also allow the community to see which event type might be best for spreading positive autism awareness.

Similarly. the change in polarity will be aggregated by the day of the week and time of the day. When the events are aggregated by day of the week, the combined events will be treated regularly as the combinations consist of events within a day. When the data is aggregated by the time of day, only events with time data are used, and those events are never combined as their time frame is adjusted to prevent a collision. Therefore, aggregation by the time of day will not have to worry about the combined events. Faceted boxplots will be created for the day of the week and the time of the day. The boxplots will be divided by an hour-long time frame for the time-of-day aggregation. So, that graph will contain 24 separate boxplots. For day-of-the-week aggregation, there will be one boxplot per day. These graphs will allow the community to see what time of the day and what day of the week are best for hosting events.

To further clarify the data analysis, the resulting graphs and data will reflect not just the effects of the autism awareness events but also other external factors. These external factors can not be eliminated as the information on what other factors could be involved is missing. With the aggregated results, Factors such as day of the week and time of the day will be analyzed. Beyond that, aggregating the result by event types will also help reduce the effect of other unknown events co-occurring as autism awareness events. With aggregated results, the effects of "one-time" external factors or events that did not always happen alongside the different event types should be reduced. Also, graphing a boxplot, which includes the median, with the aggregated data should help show the result and reduce some of the outliers caused by external or unknown events. The data could also reflect the lack of a strong correlation between the autism awareness event and the change in polarity. This lack of correlation would mean that the tweets might not be an excellent way to analyze the change in sentiment due to autism awareness events. However, there should be a correlation as other studies have confirmed a correlation between autism awareness events and changes in sentiment. One researcher also found a change in the volume and sentiment of autism awareness tweets with a similar analysis process [27]. Therefore, using the aggregated results along with the boxplots to reduce outlier should further improve the analysis and not have to worry about lack of correlation.

The visual information is for communicating the analysis to the organizations. For the portion of the analysis where the graphing and other visual aids are created, the data will be transferred from the Dask [19] data frame to the Rapids [17] data frame, and most of the work will be done locally. The final analysis portions cannot utilize the advantages of the partitioned Dask data frame as all the data will have to be loaded all at once for libraries like Seaborn [20] and Plotly [15] to graph the data. Therefore, for this portion of the analysis, the Rapids data frame will be used and completed locally.

4. Communication Specification

Communicating the process and results of this case study is essential to achieving the desired goal. These goals include making autism awareness events more efficient and allowing for further research into what makes autism awareness events more effective. Therefore, the communication strategies will be accustomed to the audience to ensure these goals are met.

4.1 Audience Communication Strategy

This case study aims to determine what event and promotion strategies best help the autism community promote and better bring awareness to autism. This case study is also exploratory, so the results and processes in the case study must be structured and appropriately communicated for further research. With that, the results need to be relayed to two groups: autism organizations that plan the awareness events and the autism research community.

4.1.1 Autism Organizations

Autism organizations include groups such as the Autism Society [3]. The information that needs to be relayed to this group includes the overall result of the research. The goal is to make these organizations plan events most beneficial to autism awareness. Therefore, they need to be presented with information such as which events should be given more funding and which events should be avoided. They will also be presented with the overall change in sentiment with time. The effects of other factors such as time and day of the week will also be explained. This information should hopefully result in better use of donations and other financial resources provided to these organizations. Further, this should also help make these organizations more efficient in spreading information about autism to counteract the vast amount of online misinformation.

4.1.2 Autism Research Community

This case study will be exploratory, so it will only determine which events in the past have led to significant positive sentiment toward autism. It would require further in-depth research to determine what factors about the events are affecting the sentiment. For in-depth research, the process and results from the case study must be appropriately communicated to the autism research community. Clear communication will require that the deliverables, which should consist of the research paper and any resource used in the research, presented to the research community be thorough and detailed. The details should allow others to replicate and expand from the case study.

4.2 Methods of communication

Given that each audience will require the information presented at various levels of complexity and detail, there will be three different deliverables. First, autism organizations require a high-level presentation. Second, the researchers will require more details, requiring a written report and a code repository.

4.2.1 Presentation

Autism organizations do not need an in-depth explanation of the case study process to utilize the result. Therefore, the autism organization will get a high-level explanation of the result in a presentation. This presentation will communicate the case study analysis results and critical factors. The key factors would be high-level information on where the data came from and the analysis process. Overall, the presentation should be more visual with graphs to compare sentiment changes during the different past events. The presentation should result in them understanding which events in the past have worked the best and which ones are a waste of time and resources. The rest of the responsibilities would lie in the organizations taking that information and using it in planning future events.

4.2.2 Written Report

The report will be highly detailed with information on how the case study will be conducted and the analysis thoroughly explained. This deliverable aims to update the research community with information on the current progress and determine where to build upon this research. In addition, this report should allow for the study to be replicated to confirm the accuracy of the results.

4.2.3 Code Repository

The repository contains all the code created during the study with detailed documentation. The main goal of this deliverable is to allow other researchers to understand the exact process better and allow replication of the research. Giving researchers the code will also allow them to optimize the methodology and process. The repository will also include documentation on the libraries and library versions used to run the code. In addition, the data for the automatic labeler and models created during the study will also be stored. The overall Twitter data will not be included in the repository as it is already organized and readily available in the Twitter API [25]. However, aggregated data from the analysis will be cached in the repository. The repository should help researchers assess the validity of the case study and expand on it to further research.

4.3 Standards

For the communication to be effective, specific standards will be utilized when writing the deliverables and the code.

4.3.1 Writing Standard

The case study will create the reports using the IEEE writing standards [4]. The IEEE writing standard best fits research where programming is involved. Given that the case study will need to explain the programming process and tools used thoroughly, this standard is the best fitting. In addition, the writing standard allows referring to code snippets and much more to help explain both the process and the code. Overall, this writing standard will allow the information to be efficiently relayed to the researchers.

4.3.2 Coding Standard

The coding standard will follow the PEP 8 standards [16] created for the Python program language. Utilizing PEP 8 standards will ensure the code is consistent and clear. In addition, the standard will allow the research community to better understand and utilize the code in the future.

4.4 Tools

This study utilizes different tools for different parts of the case study. The overall coding process will be done with Python, utilizing parallel processing libraries such as Dask [19] and Rapids [17]. Those libraries offer optimizations for working with vast amounts of data. The Twitter API [25] will be used to retrieve the data. This interface allows for historical data to be retried from Twitter directly into the python program. The vast amount of data will be handled by Dask in parallel. GitHub will be used to version and publish the code. The Microsoft Office Suite, including PowerPoint and Word, will be used for the reporting process.

5. Ethical, Legal, and Privacy Concerns

Twitter data will be the biggest concern for the legality, ethics, and privacy. However, the second dataset possesses zero concern as the information is meant for public consumption.

5.1 Concerns with Social Media Data

Twitter disclosed that the tweets accessible in the Twitter interface [25] are public and shared with the user's consent. For private accounts, their tweets are not accessible from the interface. The main concern with using tweets are the images, videos, and content other than the tweet string that might fall under other restrictions. Those restrictions will not matter because the case study will only use the tweet string, creation date, and language data. The contents of the tweet other than the text will be avoided.

The limited data use helps reduce some ethical, legal, and privacy concerns, but other factors must be considered. Legal concerns are eliminated by the statement given by twitter [25] allowing the researchers to access the data. As the data was accessed with full permission through a portal provided by Twitter, the legality of accessing the data will not be of concern. Privacy and ethical concern require further thoughtful analysis of what data will be used and how it will be used. Privacy concerns are eliminated once the tweet strings are cleaned during the clean-up process. The clean-up process removes any Twitter handles or usernames in the string. Further, no research or analysis is done to determine how specific people's sentiments change over time. Additionally, no hashtags are used to avoid connecting the data to a specific group of people. The sanitization process prevents any of the tweets from being connected back to a user or a conversation. Further, the tweets will be aggregated in the final result preventing any final analysis from being connected to a user. The remaining concern is with ethics and how the data will be used. Even if the data is public, the users might not want their tweets used for immoral research. Given that this research is for helping the autism community without any exploitation of the data, the ethical concern is further reduced. Additionally, the information extracted from the data is shared at no cost, which should further help reduce ethical concerns. Eliminating all ethical issues would require explicit consent from each user for having their data used for the research. However, that is not possible given the vast number of tweets. Overall, ethical, legal, and privacy concerns are eliminated or reduced.

6. Resources

In order to accomplish all the tasks above, the study will require resources beyond just the tools stated above. These resources include personnel, software, and hardware. All of these resources have their specific tasks and purpose during the study.

6.1 Personnel

Given the vast amount of work involved in the study, dividing the work among different people will make the work easier and more efficient for everyone. Therefore, the study will require two data scientists, two data labelers, a graphic designer, and a manager to organize everyone.

6.1.1 Manager

The team size is large enough that it would require personnel to organize the team. Therefore, a manager will help keep people on track and organize/assign tasks. A developmental framework will not be utilized as the developmental process in the case study is straightforward. The manager will take all the responsibility and manage the tasks and the case study timeline. The manager will also ensure everyone has the right resources to accomplish their tasks.

6.1.2 Data Scientist

The role of the data scientist covers most of the case study. Their role ranges from retrieving the data to authoring a report. This range includes retrieving the data from the sources, cleaning up the data and preparing it for analysis with event type label confirmation, completing the analysis, creating basic visuals for the analysis, and communicating the process and results to others in the team. Additionally, they will need to author the report, ensuring that the research is explained thoroughly, allowing others to repeat and confirm the study. As the role of the data scientist entails a vast amount of work, two data scientists will be required. With two data scientists, their differing experiences with different tools used in the research will also help extract more information from the analysis making the results betters. Having two data scientists also allows them to cross-examine their work and ensure everything is done correctly, making the analysis and work more accurate.

6.1.3 Data Labeler

Data labelers will be necessary during the data retrieval and data clean-up process. As both the datasets require manually adjusting the labels once automatically labeled, the labelers will help validate the labels. The labelers will go through the different events, and the label conflicted tweets. If they find an error or believe that the event type must be adjusted for the event dataset, they will adjust the label by adding the new label and reasoning to two separate columns. The data scientist will discuss and confirm the label if there are conflicts or when one labeler mends while another does not and situations where the label does not agree. They will also include their reasoning. This method ensures that the study is repeatable and ensures consistency. Similarly, they will also resolve the conflicts between VADER [26] and TextBlob [21] polarity scores. For each row with conflicting polarity, the labelers will pick one of the two polarities based on their reasoning and store both the reasoning and pick. Again, the data scientists will only be involved if conflicts exist between the labeler's decisions. They will resolve those conflicts by discussing with each other and picking one of the options and storing the reasoning. Both the processes are further explained in the data analysis section. Their role is crucial as they help reduce the number of errors the data scientist will have to spend time on.

6.1.4 Graphic Designer

Another goal of the case study is to convey the results to different organizations. The role of the graphic designer will be to create appealing visuals to help communicate the results to the organizations. The graphic designer will aim to capture the audience's interest and ensure the correct information is conveyed. Their tasks will consist of creating the presentation with heavy use of visuals. The visuals will build upon what the data scientist extracted from the analysis. Again, the data scientists are responsible for ensuring the designers have adequate information to create the visuals. Finally, the data scientists will be responsible for presenting the visuals to the organizations.

6.2 Software

Each of the team members will require different software to complete their tasks. Starting with the manager, they will use the primary tool or software Trello [23] and Gmail. These are both web apps. Trello does require licensing per user and will be used by all personnel. Trello will be used for managing tasks and timelines, while the Gmail service will be used to communicate.

The data scientists' work will use Python with different libraries such as JupyterLab to accomplish most of their work. Libraries such as NLTK [11], Pandas [13], Numpy [12], TextBlob [21], Scikit learn [9], and Seaborn [20] will also be used for data clean-up, analyzing, and creating visuals. As stated earlier, Dask and Rapids [17] libraries will be used for parallel computing. Python and all the libraries that will be used will not require licensing or any purchases. Also, they will not require a database because the Twitter API [21] organizes the data. In order to ensure work quality and track progress, a repository such as GitHub [29] will be used to manage their work. In addition, a team repository will be used free of charge. For the report at the end of the study, they will utilize the Microsoft Office suite [10], which does require a license. Their primary software will be Microsoft Word, which will be used for authoring the paper. References will be managed using tools inside Word. Data labelers will utilize tools that the data scientists create to help adjust the labels. Therefore, they will not require any specific software other than what everyone else will be using.

The graphic designers will create graphics other than those created by the data scientists using Seaborn. They will use Adobe software [2] such as Photo Illustrator and Photoshop for graphics other than the data visuals. In addition, they will use software such as Microsoft Power BI [5] to create data visuals. Overall, each team member other will have their specific and shared software.

6.3 Hardware

Each team member will require a laptop to work. The laptop belonging to the data scientist and graphic designer will need specifications that match a workstation. Their laptops will also need a dedicated Nvidia graphics card. The other team members' laptops will not require such high-end specifications. Other than the laptops, the data scientist will also require access to cloud computing. As the data wrangling and the analysis process will be using Dask [19], the Dask workers will be distributed among multiple compute units. The Dask client on the data scientists' computer will connect to these workers to accomplish tasks in parallel. The Dask workers will require multiple Amazon Web Services (AWS) Elastic Compute

Cloud (EC2) instances with CPU and memory specifications matching those of the workstation laptops. The data will be cached in the laptops allowing the EC2 instances to run with lower storage capacity. The Dask library does not contain all the required functionality for the analysis. Therefore, some of the analysis and data processing will be done locally, requiring a high-end specification laptop. These local processes will be done using Rapids [17] which requires the graphics card. For the graphics designer, a high-end laptop with a graphics card would speed up and optimize the overall workflow.

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